MILITARY SPECIFICATION

CANNON, 105MM HOWITZER: M205

This specification is approved for use within the US Army Armament Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification supports the acquisition of the Cannon, 105MM, M205 for use with the Howitzer, Light, Towed, 105MM: M204. The Cannon consists of a tube and a breech mechanism assembly.

1.2 This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-13931	-	Cannon, General Specification for
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-H-63192(AR)	-	Howitzer, Light, Towed, 105MM: M204

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1015

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STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1189	-	Standard Department of Defense Bar Code Symbology
MIL-STD-1235	-	Single and Multi-Level Continuous Sampling Procedures and Table for Inspection by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government document, drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

OTHER GOVERNMENT DOCUMENT

U.S. Army Armament, Munitions and Chemical Command

Master LQAP 11579150	- Master List of Quality Assurance Provision for Cannon, Provision for Cannon, 105MM Howitzer: M205

DRAWINGS

U.S. Army Armament, Munitions and Chemical Command

11579150	- Cannon, 105MM Howitzer: M205
P11579150	- Packaging Data Sheet, Cannon 105MM Howitzer: M205

PUBLICATION

U.S. Army Armament, Munitions and Chemical Command

QAP-APPENDIX-WVA	- General Quality Assurance Provisions

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents may also be available from libraries or other informational services).

2.3 Order of precedence. The applicable contract contains an order of precedence clause wherein the specifications are listed last. With respect to this specification, the order of precedence is drawing requirements, followed by the text of this specification, then the other referenced specifications and standards. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer of the design activity.

3.2 General requirements. The cannon and component parts shall be in accordance with the applicable contract, the applicable drawings (11579150 and P11579150), the requirements of this specification, the provisions of MIL-C-13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2.1 First Article - Initial Production Testing (FA-IPT) Requirements. The requirements for First Article - Initial Production Testing are collected and included in Appendix A. When specified, these FA-IPT requirements are mandatory for compliance.

3.3 Functional characteristics. The cannon shall perform the firing, pressure sustaining, and projectile launching functions of the Howitzer, Light, Towed 105mm: M204. The breech mechanism, firing mechanism, and other moving parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. When the operating lever is unlatched and rotated clockwise (rearward), it shall rotate the crank clockwise. The rotation of the operating lever shall cock the firing mechanism and cause the horizontally sliding breechblock to move to the right, and open the breech. The extractor shall eject the empty cartridge case from the breech chamber. When the operating lever is rotated counterclockwise, the breechblock shall move to the left, and close the breech.

3.3.2 Firing mechanism group. During the first thirty-five degrees clockwise rotation of the operating lever, the crank shall rotate the retractor cam counterclockwise. The rotation of the retractor cam shall cause the retractor to force the firing pin rearward past the sear and the firing pin spring shall compress. Rearward movement of the firing plunger shall pivot the trigger and the sear, which shall rotate and release the firing pin. The rotation of the sear and release of the firing pin shall cause the firing pin spring to expand and drive the firing pin forward to strike and fire the cartridge primer.

3.3.2.1 Firing plunger reset position. The firing plunger reset position is an

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indication of the distance that the firing plunger is cammed forward as the breech mechanism is opened. This position shall be measured as the minimum recessed distance from the front face of the breech ring to the front end of the firing plunger. This distance shall be not less than 0.055 inch nor more than 0.110 inch.

3.3.2.2 Firing plunger fire position. The firing plunger fire position is an indication of the location of the firing plunger just as the firing pin is released by the sear (breech mechanism closed). This position shall be measured as the recessed distance from the front face of the breech ring to the front end of the firing plunger. This distance shall be not less than 0.330 inch nor more than 0.480 inch.

3.3.2.3 Firing plunger final position. The firing plunger final position is an indication of its maximum possible rearward travel (breech mechanism closed). This position shall be measured as the recessed distance from the front face of the breech ring to the front end of the firing plunger. This distance shall be not less than 0.540 inch nor more than 0.580 inch.

3.3.2.4 Firing pin protrusion. The firing pin protrusion measured from the front face of the breechblock shall be not less than 0.130 inch nor more than 0.145 inch in the fired position.

3.3.2.5 Firing pin over-travel. Firing pin over-travel is an indication of the maximum amount of firing pin travel beyond the sear as the breech mechanism is opened. Over-travel shall be not less than 0.145 inch nor more than 0.278 inch.

3.3.3 Headspace. Headspace, the distance between the front face of the closed and locked breechblock and the rear face of the tube, shall be not less than 0.190 inch nor more than 0.200 inch.

3.4 Reliability, availability, maintainability and durability (RAM-D). The RAM-D requirements of the M204 Howitzer system are stated in MIL-H-63192(AR). Since the availability requirements are specified for the M204 Howitzer system only, this attribute cannot be apportioned to the cannon. The requirements for Reliability, Maintainability, and Durability are collected and included in Appendix B. When specified, these requirements are mandatory for compliance.

3.5 Performance.

3.5.1 High pressure resistance. The cannon and spare tubes shall be capable of withstanding a chamber proof pressure of 56,000 + 1,000 psi within its operating temperature range (see 30.3 of Appendix A) when tested as specified in 4.7.6.

3.5.2 Material soundness. All parts shall comply with soundness requirements specified on the applicable drawings.

3.5.3 Bore enlargement. The bore diameters, measured across the rifled lands after proof firing no more than eight rounds shall not exceed the before firing diameters by more than the amounts shown in TABLE I.

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TABLE I. Bore enlargement.

Distance forward of the Commencement of rifling	0.10 inch	1 inch	1 cal.	2 cals to muzzle
Amount (inches)	0.008	0.008	0.006	0.003

NOTE: Interpolate to determine the amount between given distances.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance conform to all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. First article initial production test (see 4.5).

c. RAM-D Inspection (see 4.6).

d. Quality conformance inspection (see 4.7).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test document, applicable paragraph(s) in the specification, or in the applicable drawing.

4.4 First article inspection. Unless otherwise specified in the contract (see 6.2.1), first article cannon and specified parts thereof shall be submitted for inspection in accordance with QAP-APPENDIX-WVA and applicable item QAPs. First article cannon and parts thereof shall be representative of the production

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processes to be used during quantity production. Quality Assurance Provisions (QAPs) are listed in Master LQAP 11579150. First article quantities are specified in the applicable item QAPs.

4.5 First Article Initial Production Test (FA-IPT). The requirements tests, and test methods for FA-IPT are contained in Appendix A.

4.6 Reliability, Maintainability, and Durability. The requirements, tests, and test methods for reliability, maintainability, and durability are contained in Appendix B.

4.7 Quality conformance inspection. Quality conformance inspection shall be as specified in Table II.

4.7.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance to the requirements of this specification, to MIL-C-13931, QAPs listed in Master LQAP 11579150 and QAP-APPENDIX-WVA and any additional requirements listed in the contract (see 6.2.1).

4.7.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

TABLE II. Quality conformance inspection.

Inspection	Requirement Paragraph	Test paragraph	Test method paragraph
1. Firing Plunger a. reset position b. fire position c. final position	 3.3.2.1 3.3.2.2 3.3.2.3	 4.7.4.1 4.7.4.2 4.7.4.3	 4.7.7.1.1 4.7.7.1.2 4.7.7.1.3
2. Firing pin a. protrusion b. over-travel	 3.3.2.4 3.3.2.5	 4.7.4.4 4.7.4.5	 4.7.7.1.4 4.7.7.1.5
3. Headspace	3.3.3	4.7.4.6	4.7.7.1.6
4. High Pressure resistance	3.5.1	4.7.6	4.7.7.3.1 4.7.7.3.2
5. Material soundness	3.5.2	4.7.4.7	4.7.7.1.7
6. Bore Enlargement	3.5.3	4.7.6	4.7.7.3.1
7. Packaging	5.	4.7.4.8	4.7.4.8
8. Breech mechanism and firing mechanism group	3.3, 3.3.1 3.3.2	4.7.5.1	4.7.7.2.1

4.7.3 Inspection equipment.

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4.7.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1) responsibility for acquisition, maintenance, and disposition of inspection equipment prescribed in applicable item QAP's, shall be in accordance with MIL-I-45607.

4.7.3.2 Accuracy of commercial inspection equipment. When commercial and modified commercial inspection equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.7.3.2.1 Accuracy of proof pressure measuring devices. The devices for measuring proof firing pressures shall have the following accuracy:

Device	Maximum error of measurement (%)
copper crusher gages	± 3 of reading
pressure transducers	± 2 of reading

4.7.4 Quality conformance examinations.

4.7.4.1 Firing plunger reset position. The firing plunger reset position shall be measured using the examination method of 4.7.7.1.1. Cannon failing to comply with the requirements of 3.3.2.1 shall be cause for rejection.

4.7.4.2 Firing plunger fire position. The firing plunger fire position shall be measured using the examination method of 4.7.7.1.2. Cannon failing to comply with the requirements of 3.3.2.2 shall be cause for rejection.

4.7.4.3 Firing plunger final position. The firing plunger final position shall be measured using the examination method of 4.7.7.1.3. Cannon failing to comply with the requirements of 3.3.2.3 shall be cause for rejection.

4.7.4.4 Firing pin protrusion. The firing pin protrusion shall be measured using the examination method of 4.7.7.1.4. Cannon failing to comply with the requirements of 3.3.2.4 shall be cause for rejection.

4.7.4.5 Firing pin over-travel. The firing pin over-travel shall be measured using the examination method of 4.7.7.1.5. Cannon failing to comply with the requirements of 3.3.2.5 shall be cause for rejection.

4.7.4.6 Headspace. Headspace shall be measured using the examination method of 4.7.7.1.6. Cannon failing to comply with the requirements of 3.3.3 shall be cause for rejection.

4.7.4.7 Material soundness.

4.7.4.7.1 Borescope and Magnetic Particle. All surfaces of each cannon tube, breech ring, and breechblock shall be magnetic particle inspected using the methods of 4.7.7.1.7.1 and 4.7.7.1.7.2. Failure to comply with requirements of 3.5.2 shall be cause for rejection.

4.7.4.8 Packaging inspection. Unless otherwise specified, inspection to

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determine compliance with packaging requirements of Section 5 and the applicable packaging data sheet for the level designated in the contract shall be as specified in MIL-P-14232.

4.7.5 Quality conformance functional inspections.

4.7.5.1 Breech mechanism and firing mechanism group. The breech mechanism and the firing mechanism shall be inspected using the method of 4.7.7.2.1. Cannon failing to comply with the requirements of 3.3, 3.3.1, and 3.3.2 shall be rejected.

4.7.6 Proof acceptance. Cannon shall be proof accepted through proof firing, simulation tests, or both. Cannon and spare tubes to be proof fired shall be tested using the method of 4.7.7.3.1. Failure to comply with the requirements of 3.5.1, 3.5.2, and 3.5.3 shall be cause for rejection. Cannon breech mechanisms to be simulation tested shall be tested using the method of 4.7.7.3.2. Failure to comply with the requirements of 3.5.1 and 3.5.2 shall be cause for rejection. Proof firing shall be done on a sampling basis and shall be conducted as follows:

a. Using sampling plan CSP-1 of MIL-STD-1235 and an AQL of 1.0, the Product Assurance Directorate at Watervliet Arsenal will designate the appropriate initial quantity (i) and sample frequency (f) (see 6.2.1.j.) in the contract.

b. The first "i" quantity of cannon or spare tubes produced in the particular buy shall be proof fired. When the "i" quantity has been satisfied, sampling shall be put into effect using the "f" sampling frequency. All cannons not proof fired shall have the breech mechanisms simulation proof tested in the Product Assurance weapon test facility at Watervliet Arsenal.

c. Successive production buys shall not require "i" quantities. However, if the time lag between two successive buys exceeds one year, or if a new cannon or tube contractor is utilized, a new "i" quantity shall be proof fired.

d. Proof fired cannon and spare tubes, and simulation tested breech mechanisms shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data) as specified by Watervliet Arsenal Product Assurance Directorate.

4.7.7 Methods of Inspection.

4.7.7.1 Examination methods.

4.7.7.1.1 Firing plunger reset position. The firing plunger reset position shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Government approved equivalent gage.

4.7.7.1.2 Firing plunger fire position. Firing plunger fire position shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Governement approved equivalent gage.

4.7.7.1.3 Firing plunger final position. Firing plunger final position shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Government approved equivalent gage.

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4.7.7.1.4 Firing pin protrusion. Firing pin protrusion shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Government approved equivalent gage.

4.7.7.1.5 Firing pin over-travel. Firing pin over-travel shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Government approved equivalent gage.

4.7.7.1.6 Headspace. Headspace shall be gaged on each cannon with the applicable gage referenced on the applicable item QAP or with Government approved equivalent gage.

4.7.7.1.7 Material soundness.

4.7.7.1.7.1 Borescope examination. The inner surface of each tube shall be examined with a black-light borescope to assure conformance with applicable drawings and specifications.

4.7.7.1.7.2 Magnetic Particle Examination. The outer surface of each tube, all surfaces of each breechblock and each breech ring, shall be magnetic particle examined in accordance with applicable item QAPs.

4.7.7.2 Functional inspection methods.

4.7.7.2.1 Breech mechanism and firing mechanism group functional inspection. The inspection shall be done either manually or through the use of a functional tester. A minimum of five complete cycles of breech operation shall be performed on each breech mechanism and firing mechanism group. An empty case cartridge, M14B3E1, shall be inserted into the chamber for each of the five cycles. When the breech is opened, the extractor shall eject the empty cartridge case from the breech chamber. For each of the cycles, the firing plunger shall be activated upon closing the breech to assure that the firing pin operates as required.

4.7.7.3 Test Methods.

4.7.7.3.1 Proof firing test. Proof firing shall be conducted at a Government specified test site. The procedure shall be as follows:

a. Each cannon or spare tube submitted shall be proof fired once at 56,000 + 1,000 psi.

b. A facilities mount may be used if necessary.

c. Chamber pressure shall be measured (see 4.7.3.2.1) and recorded.

d. Bore enlargement shall be checked with a star gage, air gage, electronic gage, or other Government approved gage.

e. Cannon shall be borescoped (see 4.7.7.1.7.1) and magnetic particle inspected (see 4.7.7.1.7.2).

f. Cannon found to be defective at the proving ground, as a result of a

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critical failure, shall be reported to the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189 and Benet Weapons Lab, Watervliet Arsenal, Watervliet, NY 12189 within one working day. The test agency shall forward an Equipment Performance Report (EPR), incident to the deficiency within three working days.

4.7.7.3.2 Breech mechanism simulation proof test. All cannon not proof fired at a proving ground shall have the breech mechanism tested in accordance with the following procedure:

a. Each breech mechanism shall be tested by means of a suitable hydrostatic pressure testing equipment for one application of an internal overpressure (proof pressure) of 56,000 + 1,000 psi. Proof pressure shall be immediately released upon attainment. Cycle time to achieve proof pressure and release shall not exceed one minute.

(1) When difficulty is encountered in attaining the test pressure the number of attempts in which 10 percent or less of the test pressure is attained shall not be counted. However, the number of attempts in which over 10 percent of the test pressure is attained shall not exceed five.

(2) If the cause of difficulty is not identified and corrected within five pressure cycles, the breech mechanism, as part of the cannon, shall be proof fired in accordance with 4.7.7.3.1.

(3) The peak pressure shall be recorded for each test pressure cycle and attached to the inspection records for the particular breech mechanism undergoing test.

b. After pressure testing, each mechanism shall be examined for looseness or interference of fit or function.

c. The breech mechanism shall be disassembled and examined for material soundness by magnetic particle inspection, conducted in accordance with specifications, procedures, and acceptance criteria as specified on applicable drawings.

5. PACKAGING

5.1 Cannon - Levels A, B and C. Preservation, packaging and packing of the cannon shall be in accordance with Packaging Data Sheet 11579150 for Cannon, 105MM Howitzer M205 for the level of protection specified in the contract (see 6.2.1).

5.2 Repair parts.

5.2.1 Levels A, B, and C. Repair parts preservation, packaging and packing shall be in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2.1).

5.2.2 Level X-industrial packaging. Repair parts preservation, packaging, and packing shall be in accordance with ASTM D 3951 when Level X protection is specified in the contract (see 6.2.1).

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5.3 Marking.

5.3.1 Levels A, B and C. Marking of unit packages and exterior shipping containers shall be in accordance with Packaging Data Sheet 11579150 for Cannon, 105MM Howitzer M205 for the level of protection specified in the contract (see 6.2.1).

5.3.2 Level X - Industrial. Marking shall be in accordance with ASTM D 3951.

5.3.3 Bar code marking. When specified, (see 6.2.1.o) bar code marking shall be in accordance MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended Use. The cannon specified herein is intended to be used in the 105MM M204 Howitzer system.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number and date of this specification.

b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article (FA), first article initial production test (FA-IPT), or RAM-D tests, by the contractor (see 3.1, 3.2.1, and 3.4).

c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

d. All inspection and test data that is required.

e. At whose expense a retest may be performed.

f. Serial numbers for the cannon and tube assemblies.

g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.7.3.1).

h. Availability of inspection equipment from the Government.

i. Extent of contractor's responsibility for Government-furnished and for contractor-required final inspection equipment.

j. Sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f); (see 4.7.6.a).

k. Under what conditions simulated proof testing may be performed, the test equipment available, and the procedures to be followed.

l. When applicable, the sampling plan for the actual firing of tubes.

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m. The levels of preservation, packaging, and packing required for cannon (see 5.1) and repair parts (see 5.2).

n. List of components that have been identified as reliability sensitive (see 30.1 of Appendix B).

o. When applicable, bar code marking requirements (see 5.3.3).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2.2.1 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.1	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120, or as directed by the Contracting Officer).

6.3 The contract should specify the application of either MIL-Q-9858 or MIL-I-45208, as appropriate.

6.4 The contract should specify the application of MIL-I-45607 and MIL-STD-45662 when warranted.

6.5 Subject term (keyword) listing.

Cannon
Howitzer

Custodian:
Army-AR

Preparing Activity
Army-AR
Project No. 1015-A267

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APPENDIX A

FIRST ARTICLE-INITIAL PRODUCTION TESTING REQUIREMENTS FOR CANNON: 105MM M205

10. SCOPE

10.1 Scope. This Appendix details all requirements and inspection methods necessary for First Article - Initial Production Testing (FA-IPT). This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this Appendix.

30. REQUIREMENTS

30.1 FA-IPT testing. FA-IPT tests shall be conducted on cannon or critical assemblies or parts thereof when:

a. a new contractor is utilized,

b. a time lapse of one year has occurred since the item was last produced by the contractor,

c. significant changes have been made in a manufacturing process or material,

d. a design change is made.

If the cannon or critical assemblies or parts thereof fail FA-IPT testing, then failure analysis shall be invoked (see 40.4).

30.2 Proof Firing Test. The requirements for the proof firing are contained in 3.5.1, 3.5.2, and 3.5.3 of MIL-C-45978.

30.3 Climatic. The cannon shall function satisfactorily within the temperature range of +125 degrees F to -50 degrees F. Satisfactory function shall be defined as the breech closing on loading a round, firing of the round and ejection of the empty cartridge case with the breech remaining open after ejection. In addition, the breech must perform the firing and ejection functions without damage to the breech components.

30.4 RAM-D. The requirements for RAM-D are contained in Appendix B of MIL-C-45978. When specified, these RAM-D requirements shall be met during FA-IPT. (Appendix A of MIL-C-45978).

40. INSPECTION METHODS

40.1 First Article Initial Production Test (FA-IPT). When specified in the contract (see 6.2.1), first article cannon shall be subjected to first article initial production testing (FA-IPT) by a Government activity to assure compliance with the requirements of this specification and the contract.

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40.1.1 Sample. Two first article cannon shall be randomly selected from the first month's production. First Article Cannon and parts thereof shall be representative of the production processes to be used during quantity production. When specified, two additional cannon shall be included for RAM-D testing.

40.1.2 Inspection Provisions. First Article Cannon shall be subjected to normal inspection per applicable item QAP's. Quality Conformance Inspection (see 4.7) including proof acceptance (see 4.7.6), and any additional requirements listed in the contract.

40.1.2.1 Physical Configuration Audit. One cannon selected from the First Article sample shall undergo a physical configuration audit in addition to the normal inspection cited in 40.1.2.

40.1.2.2 Proof Acceptance Through Proof Firing. First Article Cannon shall have successfully passed proof acceptance (see 4.7.6) prior to commencement of First Article Initial Production Testing.

40.1.3 Tests. Unless otherwise specified herein or in the contract, first article cannon tests listed in this specification shall be performed in a fully operational Howitzer, Light, Towed, 105mm: M204, and shall not normally be integrated with other tests. Integration with Operational Testing by the user is allowable provided all components of the system conform to technical requirements.

40.2 Tests.

40.2.1 Proof Firing Test. The test method for the proof firing are contained in 4.7.7.3.1 of MIL-C-45978.

40.2.2 Climatic Test. A climatic test shall be conducted on two cannon using the test method of 40.3.1. If the requirements of 30.3 are not complied with, acceptance shall be deferred and 40.4 shall apply.

40.2.3 RAM-D Tests. The tests to verify the RAM-D requirements are contained in Appendix B of MIL-C-45978.

40.3 Test methods.

40.3.1 Climatic test. A climatic test shall be conducted on two cannon with both the cannon and ammunition conditioned to the temperature specified in TABLE III. Ammunition used in this test shall be inert loaded Cartridge M760, Zone 8. Borescope (see 4.7.7.1.7.1) and magnetic particle (see 4.7.7.1.7.2) examinations shall be conducted on cannon after firing in accordance with TABLE III. Facilities components may be used if necessary.

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TABLE III. Climatic test schedule.

Rounds	Temperature of Cannon and Ammunition
5	+125°F
5	- 25°F
5	- 50°F

40.4 Failure analysis. The contractor shall assist the Government in the failure analysis procedure by performing a dimensional, physical and visual examination of the FA-IPT components which are suspected to have caused the failure. Results shall be submitted to the Government for a failure cause determination. If it is determined that failure occurred because of deviation from specified dimensions, physical properties, or both, the contractor shall accomplish, as applicable, the following actions:

a. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defects in future production.

b. Examine completely assembled cannon and partially assembled cannon and components to ensure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

c. Submit the results of the failure analysis and the corrective actions taken to the Government for evaluation and approval of retest or continuation of test.

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APPENDIX B

RELIABILITY, MAINTAINABILITY, DURABILITY REQUIREMENTS

AND RELIABILITY SCORING CRITERIA FOR CANNON: 105MM M205

10. SCOPE

10.1 Scope. This Appendix details all requirements and inspection methods necessary for Reliability, Maintainability and Durability. This Appendix also provides Reliability assessors guidelines for the consistent classification of failure for assessing the Reliability requirements of the Cannon: 105MM M205. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

STANDARDS

MILITARY

MIL-STD-721 - Definitions of terms for Reliability and Maintainability

MIL-STD-882 - System Safety Program Requirements

PUBLICATIONS

"XM204 Howitzer Scoring Criteria Plan" dated May 1973.

30. REQUIREMENTS

30.1 Reliability. The cannon as a component of a fully operational M204 Howitzer system, shall have a minimum acceptable value (MAV) of 1200 equivalent full charge (EFC) mean rounds between failure (MRBF) with a confidence level of 80%. Components most impacting cannon MRBF will be identified and designated as reliability sensitive in the contract or purchase order (see 6.2.1). To insure cannon reliability, special attention needs to be focused on these critical components. Emphasis should be placed on stringent component quality control and supplemental component tests in production.

30.2 Maintainability. The cannon parts and components shall have the following maintainabilities:

a. Tube: Cleaning and lubricating the tube after firing, in accordance with applicable maintenance procedures, to prevent corrosion shall be the only maintenance required.

b. Breech: Service of the breech shall be carried out by the crew and shall be limited to minor maintenance and cleanup. Disassembly, cleaning and reassembly

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of the breechblock, firing mechanism, and associated operating linkage shall be carried out by the crew within 1.5 man-hours. Repair shall be carried out at unit (company or battalion) level.

30.3 Durability. The durability of the M204 Howitzer system as it pertains to the cannon shall be:

30.3.1 Fatigue life. The following are the required minimum durabilities for the tube and breech mechanism:

a. Tube - 7,500 EFC rounds.

b. Breech mechanism - 22,500 EFC rounds.

30.3.2 Tube wear life. The required minimum tube wear has not been apportioned to the tube. Data from future firings shall be analyzed to establish the tube wear requirement.

40. INSPECTION METHODS

40.1 RAM-D Test. RAM-D testing shall be performed at the completion of the FA-IPT test specified in Appendix A when:

a. There is a new producer or a change in producers.

b. Significant changes in the manufacturing processes have occurred.

c. Significant changes in the Technical Data Packages (TDP) have occurred.

d. Sufficient incidents have occurred with fielded systems to indicate that a RAM-D problem exists.

e. When specified in the contract.

NOTE: When RAM-D testing is required under paragraphs b, c, d, or e, the full RAM-D testing or a portion of the RAM-D testing may be specified by the engineering or product assurance activities. The scope of the testing shall be at the discretion of the engineering and product assurance activities at a level appropriate to the changes or incidents involved. When the change or the incident relates only to a specific component, then component level testing may be substituted for RAM-D testing at the discretion of the engineering and product assurance activities. Simulation testing may be utilized if applicable.

40.1.1 RAM-D Sample. The RAM-D test shall be performed on four cannon using the methods specified in 40.1.2. When RAM-D testing is to be performed, the sample shall consist of the two cannon that completed FA-IPT testing and two additional cannon.

40.1.2 RAM-D Tests. At the completion of FA-IPT testing (see Appendix A), the RAM-D test shall be performed.

40.1.2.1 Reliability Test. The combined test (tube wear and reliability) shall

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consist of firing 5 sequences, each sequence consisting of firing 96 rounds from each of 4 cannon. The type of round fired shall be as specified in Table IV.

40.1.2.2 Cannon reliability demonstration. To demonstrate that the reliability requirement has been satisfied, the tests specified in Table IV shall be performed without a reliability failure. Failure scoring criteria for assessing reliability requirements shall be in accordance with 50.2 of Appendix B. If a failure occurs during testing, the failure analysis procedures outlined in 40.2 of Appendix B shall apply.

TABLE IV. RAM-D Test. *, **

SEQ.	CANNON REQ'D	ROUNDS PER CANNON	M760 ZONE 8 (TOTAL)	M760 ZONE 8 (CUM.)
1	4	96	386	386
2	4	96	386	772
3	4	96	386	1,158
4	4	96	386	1,544
5	4	96	386	1,930

GENERAL NOTES:

a. To demonstrate that the tube wear requirement has been achieved, the tests specified in Table IV shall be completed within the tube wear requirement. Failure to do so shall require the application of the procedures outlined in 40.2.

b. Rate of fire shall not exceed 3-5 rounds every one minute.

* Borescope and measure tube ID at the beginning of each sequence.

** Magnetic particle inspect, borescope, and measure tube ID at the end of each sequence.

40.1.2.3 Cannon durability test method. The durability test shall be conducted on two cannon, essentially by hydraulic pressure cycling at the applicable service chamber pressure(s) of the cannon tube (54,000 ± 1,500 psi) and breech mechanism (breech ring and breechblock) (54,000 ± 1,500 psi). Rounds fired in other tests of Appendices A and B may be counted as cycles for the durability tests.

40.1.2.3.1 Tube durability test method. The two tubes shall be conditioned by initial firing of 250 rounds from each tube. Durability shall be demonstrated by hydraulically pressurizing the two tubes at Watervliet Arsenal or other Government approved facility. Each tube shall be cycled to destruction at pressures conforming to the pressure travel curve for service ammunition. The fatigue life

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shall be consistent with the requirement of 30.3.1.a based on a log normal failure distribution. If the requirement is not satisfied, 40.2 of Appendix B applies.

40.1.2.3.2 Breech mechanism durability test method. Durability shall be demonstrated by hydrodynamic impact testing two breech mechanisms (breech ring and breechblock) at Watervliet Arsenal or other Government approved facility. Each breech mechanism shall be cycled to destruction at 43,700 + 1,500 psi. The fatigue life shall be consistent with the requirement of 30.3.1.b based on a log normal failure distribution. If the requirement is not satisfied, 40.2 of Appendix B applies.

40.2 Failure analysis. The contractor shall assist the Government in the failure analysis procedure by performing a dimensional, physical and visual examination of the FA-IPT components which are suspected to have caused the failure. Results shall be submitted to the Government for a failure cause determination. If it is determined that failure occurred because of deviation from specified dimensions, physical properties, or both, the contractor shall accomplish, as applicable, the following actions:

a. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defects in future production.

b. Examine completely assembled cannon and partially assembled cannon and components to ensure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

c. Submit the results of the failure analysis and the corrective actions taken to the Government for evaluation and approval of retest or continuation of test.

50. RELIABILITY SCORING CRITERIA

50.1 Basis. This scoring criteria is based on the reliability scoring procedure approved by the Department of the Army. It was extracted from the "XM204 Howitzer Scoring Criteria Plan" dated May 1973. The Failure Scoring Procedure (see 50.2) provides for the use of the Failure definition (see 60.1.3) which is amplified (see 50.1.1). Failure modes (see 50.1.2), Degraded Modes of Operation (see 50.1.3) and Failure exclusions (see 50.1.1.1) pertaining to the cannon provide additional failure scoring guidelines as specified in the "XM204 Howitzer Scoring Criteria Plan" dated May 1973.

50.1.1 Failure definition amplification.

50.1.1.1 The following are not considered as reliability failures:

a. Scheduled replacement of parts before failure.

b. An incipient malfunction corrected during scheduled preventive maintenance on the part in question provided a higher level of maintenance is not necessary.

c. A malfunction resulting from not following the prescribed operational or maintenance procedures or schedule dictated by the equipment manuals.

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d. A malfunction resulting from test item abuse, unrealistic operating conditions or accident.

e. Actual or incipient malfunctions detected or corrected during initial technical inspections and incipient malfunction detected during final technical inspection.

50.1.1.2 The following are considered as reliability failures:

a. A failure detected and/or corrected during the correction of another failure provided the failures are totally unrelated.

b. Corrected incipient malfunctions not covered by 50.1.1.1 examples b and e.

c. Failure detected and/or corrected during final technical inspection.

d. Failures resulting from lack of clarity of instruction or other fault in the maintenance test package.

50.1.2 Failure modes. The following are two typical failure modes for use in assessment of mission reliability failures. These failure modes are subject to performance degradation (see 50.1.3).

1. Breechblock cannot be closed or opened.

2. Gun will not fire.

50.1.3 Degraded modes of operation. A certain amount of performance degradation must take place before a reliability failure can be assumed. Of course, if the system ceases to function, performance becomes unacceptable (i.e., non-existent); however, if performance is only degraded, then some threshold must be established to define where reliability failures begin. Allowable degradation thresholds have been established for the various performance parameters. Performance beyond these thresholds constitutes a reliability failure. As long as performance remains above the threshold or is returned above the threshold within five minute maintenance period, no reliability failure is charged. The allowable degradation thresholds are:

1. Rate of Fire - maximum	15 RPM (Rounds Per Minute)
2. Rate of Fire - sustained	3-5 RPM (Rounds Per Minute)

50.2 Failure scoring procedure. The following scheme provides the instructions for the use of the Failure Definition (see 60.1.3) for assessing reliability requirements of the cannon, 105mm Howitzer; M205 (see 50.1). These instructions assume that an incident or malfunction has occurred during test, and a record has been prepared against this action to the extent where it is necessary to determine the chargeability of the action. Incidents where the identification of a reliability failure is in doubt will be deferred to a user/developer/tester Scoring Conference. The step-by-step instructions for using the Failure Definition are as follows:

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Step 1. Record the incident information (i.e. rounds on test item and system, description of incident, nature of maintenance action required, maintenance time in hours, maintenance manhours, etc.) when a malfunction of the cannon occurs or is detected. Proceed to the next step answering the question posed and taking action according to the answer given.

Step 2. Is this a scheduled replacement of parts before failure? If yes, do not charge a failure; if no, proceed to Step 3.

Step 3. Is this a malfunction resulting from not following the prescribed operational or maintenance procedures or schedule dictated by the equipment manuals? If yes, do not charge a failure; if no, proceed to Step 4.

Step 4. Is this a malfunction resulting from test item abuse, unrealistic operating conditions or accident? If yes, do not charge a failure: if no, proceed to Step 5.

Step 5. Is this an actual or incipient malfunction detected or corrected during initial technical inspection or an incipient malfunction detected during final technical inspection? If yes, do not charge a failure; if no, proceed to Step 6.

Step 6. Is this an incipient malfunction corrected during scheduled preventive maintenance on the part in question, provided a higher level of maintenance is not necessary? If yes, do not charge a failure; if no, proceed to Step 7.

Step 7. Is this incident related to or caused by another reliability failure? If yes, do not charge a failure; if no, proceed to Step 8.

Step 8. The following statements are addressed collectively. (A) The malfunction did not and would not cause a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882. (B) The system performance is above acceptable levels (see 50.1.3). (C) The malfunction did not and would not preclude the ability to commence or the cessation of any mode of operation (e.g., load and fire at any range, charge, elevation, or direction) (see 50.1.2). If all three of the above statements are true, do not charge a reliability failure; if any are not true, proceed to Step 9.

Step 9. Is the operator/crew authorized and able to remedy the malfunction by adjustment, repair, or replacement action within five minutes, using the controls, Basic Issue Items (BII), Items Troop Installed or authorized (ITIA) and parts authorized to the crew? If the answer is yes, do not charge a reliability failure; if no, proceed to Step 10.

Step 10. If the previous 9 steps provide inconclusive evidence as to the chargeability or non-chargeability of a reliability failure, the incident should be checked against the criteria of 50.1. If categorization of the incident remains in doubt, defer classification to a user/developer/tester Scoring Conference.

60. RAM-D DEFINITIONS.

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60.1 Reliability, Availability, Maintainability, and Durability (RAM-D) terms and definitions. Reliability, maintainability and durability terms and definitions used in this specification are in accordance with MIL-STD-721 and 60.1.1.

60.1.1 Durability. Durability is the probability that an item will successfully survive its projected life, overhaul point, or rebuild point (whichever is the more appropriate durability measure for the item) without a durability failure.

60.1.2 Durability failure. A durability failure is considered to be a malfunction that precludes further operation of the item and is great enough in terms of cost, safety, or time to restore, that the item must be replaced or rebuilt.

60.1.3 Failure. For the purpose of assessing reliability, a failure is a malfunction that the operator or crew cannot or is not authorized to remedy by adjustment, repair, or replacement action within five minutes, using the controls, Basic Issue Items (BII), Items Troop Installed or Authorized (ITIA) and parts authorized to the crew, which causes or would cause:

a. inability to commence or the cessation of any mode of operation.

b. a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882.

For related malfunctions only the primary malfunction will be counted against reliability (see 50.1.1).

60.1.4 Upper pressure limit. The upper pressure limit (see 30.1) is that value of the maximum pressure which is specified in the propellant specification as the upper limit of average pressure which may be developed by an acceptable propellant in the form of propelling charges which will impart the specified muzzle velocity to the specified projectile at its standard zone weight.

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MIL-C-45978	CANNON, 105MM HOWITZER: M205

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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